Restriction Enzyme Digestion

M03 | ver 1.1 | Revised 2026-10-03

SUSTech-BIO 2026 iGEM Team

Purpose

Remove methylated parental plasmid with FastDigest DpnI, or perform sequence-directed single/double digestion with FastDigest EcoRI, BglII and MluI. DpnI template removal is a separate workflow from preparing restriction fragments for cloning or checking a construct.

Materials & Equipment

Item	Manufacturer	Identifier / specification	Use / Specification
FastDigest DpnI	Thermo Scientific	FD1703 / FD1704	Dam-methylated parental-template removal [2]
FastDigest EcoRI	Thermo Scientific	FD0274	G^AATTC; single/double digestion [3]
FastDigest BglII	Thermo Scientific	FD0084	A^GATCT; single/double digestion [4]
FastDigest MluI	Thermo Scientific	FD0564	A^CGCGT; single/double digestion [5]
10× FastDigest Buffer / Green Buffer	Thermo Scientific	B64 / B71	Choose one; colorless or with loading dyes [7]
Plasmid DNA or PCR material	Prepared sample	Sequence and concentration known	Digestion substrate
DEPC-treated water	Sangon Biotech	B501005	Laboratory water for reaction preparation [11]
Reaction tubes, filter tips, pipettes; mini centrifuge	Axygen tips; Eppendorf centrifuge; laboratory equipment	Compatible sizes	Reaction assembly and brief spin
Heat block / water bath	Laboratory equipment	37 °C; 80 °C if applicable	Digestion and heat inactivation
Crushed ice	Laboratory ice maker	Crushed ice bath	Keep enzyme tubes cold during setup

Item	Manufacturer	Identifier / specification	Use / Specification
DNA cleanup and gel system	As available / M02	Suitable for fragment size	Cleanup; M02 electrophoresis and recovery

Before You Start

These conditions apply to the confirmed FastDigest series. Catalog numbers identify reference pack sizes; check the bottle label when replacing a product. Store enzymes at -20°C and keep them on ice during setup.

Map every recognition site in the full construct, including insert, backbone and junctions. Predict fragment sizes, not only band count. A circular plasmid cut at one unique site becomes one full-length linear fragment; fragments of similar size can co-migrate.

Find the reaction conditions

FastDigest reaction conditions: temperature, time, end-flanking bases and inactivation [1]

Thermo DoubleDigest Calculator: use for conventional enzymes, not to replace FastDigest conditions [10]

Open the exact enzyme manual in Reference [2-5]. Check substrate type, buffer, temperature, incubation window, methylation sensitivity and heat inactivation before choosing a single or double digest.

Procedure

A. Design the digestion environment

1. Choose colorless FastDigest Buffer for fluorescence-based measurements or when loading dyes are unwanted. Choose Green Buffer for direct gel loading. Both support the same digestion chemistry; Green is not a separate enzyme-specific buffer. [7]

2. For purified DNA, add one-tenth of the final reaction volume as 10× buffer to obtain 1×. Calculate DNA volume V = required mass / concentration, then water = final volume - DNA - buffer - all enzymes. The buffer supplies the required ionic/pH environment, including Mg^{2+}; do not add PCR buffer, extra $MgCl_2$ or another restriction buffer by default. [2-6]

Keep the combined enzyme volume at or below 10% of the final reaction. FastDigest is dosed in µL/FDU, not interchangeable conventional units. For a scaled double digest, increase both enzymes, buffer and total volume together: doubling the 1 µg / 20 µL example gives 2 µg DNA, 2 µL of each enzyme, 4 µL 10× buffer and water to 40 µL. [4, 7]

FastDigest enzyme	37°C starting time	Heat inactivation	Longest recommended incubation
DpnI	5 min: purified methylated plasmid	80°C, 5 min	16 h
EcoRI	15 min: plasmid; 20 min: PCR DNA	80°C, 5 min	30 min
BglII	20 min: plasmid; 30 min: PCR DNA	Not heat-inactivatable	16 h
MluI	5 min: plasmid or PCR DNA	80°C, 5 min	16 h

Times use the product manuals [2-5]; the current overview [1] lists 5 min for EcoRI plasmid digestion, while the matched manual specifies 15 min, used here. Longest times are condition-dependent manufacturer limits, not routine targets. For volumes above 20 µL, allow the manual-recommended extra 3-5 min for equilibration without exceeding the shortest enzyme limit.

B. DpnI: remove parental plasmid after PCR

3. Confirm that the parent plasmid came from a dam+ host and contains GATC sites. FastDigest DpnI targets fully adenine-methylated GATC; ordinary newly amplified PCR DNA is unmethylated and should remain intact. DpnI does not remove nonspecific PCR amplicons. [2]

4. Choose one setup below. Use the purified route if the PCR mixture contains inhibitory additives or compatibility is uncertain. For direct treatment, the 30 µL recipe adapts Thermo's general crude-PCR dilution/buffer guidance to parental-template removal; PCR mixture occupies one-third of the digest. [2, 6]

Component	Purified PCR material / 20 µL	Unpurified PCR mixture / 30 µL
DNA / PCR input	V µL; conservative input up to 1 µg total DNA	10 µL completed PCR mixture
10× FastDigest Buffer	2 µL	2 µL
FastDigest DpnI	1 µL	1 µL
Nuclease-free water	(17 - V) µL	17 µL
Total	20 µL	30 µL

The crude-PCR recipe intentionally adds 2 µL, not 3 µL, of 10× buffer: PCR salts carry over. This is a specific mixed-buffer recipe, not a general 1× calculation. Do not add 10× buffer to an entire 25/50 µL PCR by eye. Split or proportionally scale this recipe, or clean up first. Colorless buffer simplifies subsequent loading-dye adjustment. [6]

5. Add water, buffer and DNA, then DpnI last; mix gently and briefly spin. Incubate the purified setup at 37°C for 5 min. For the crude-PCR adaptation, start with 15 min at 37°C and assess compatibility with the PCR premix; use cleanup if inhibition persists. Inactivate DpnI at 80°C for 5 min before a sensitive downstream step. [2, 6]

C. Single and double digestion: EcoRI, BglII and MluI

6. For cloning through restriction sites introduced by PCR primers, purify the amplicon before digestion to remove active polymerase and additives. Check the extra 5'-flanking bases required beyond the site: EcoRI 2 bp; BglII and MluI 3 bp in the FastDigest guide. Also check closely spaced sites in a double digest. [1, 3-5]

7. Assemble a single or double digest using purified DNA. Use the same one buffer for both enzymes, and add enzymes last. The PCR-DNA columns keep the manufacturer's 30 µL scale but use 3 µL buffer because PCR salts have been removed. [3-6]

Component	Plasmid single / 20 µL	Plasmid double / 20 µL	Purified PCR single / 30 µL	Purified PCR double / 30 µL
DNA	V µL; up to 1 µg	V µL; up to 1 µg	V µL; ~0.2 µg	V µL; ~0.2 µg
10× buffer	2 µL	2 µL	3 µL	3 µL
Enzyme A	1 µL	1 µL	1 µL	1 µL
Enzyme B	-	1 µL	-	1 µL
Water	(17 - V) µL	(16 - V) µL	(26 - V) µL	(25 - V) µL

Example: for 1 µg plasmid at 200 ng/µL, a 20 µL double digest contains 5 µL DNA, 2 µL 10× buffer, 1 µL of each enzyme and 11 µL water. If calculated water is negative, increase the total volume and recalculate buffer; do not omit buffer.

8. Mix gently, briefly spin and incubate at 37°C. All three enzymes share this temperature and FastDigest buffer. Use the slower enzyme's substrate-specific time, allowing for volume, while staying within the shorter maximum window. Examples: EcoRI + MluI plasmid, 15 min; BglII + MluI plasmid, 20 min. [1, 3-5]

A 30 µL EcoRI + BglII PCR digest has little margin: BglII needs 30 min, while EcoRI is limited to 30 min. If the required equilibration or extension would exceed this, digest with BglII first, purify, then perform the EcoRI digest in fresh buffer. For future enzymes with different temperatures/buffers, check both manuals and select a compatible simultaneous or sequential setup. [3, 4, 10]

9. Stop or clean up according to the enzymes present. EcoRI and MluI can be heat-inactivated at 80°C for 5 min. BglII cannot: use an appropriate DNA cleanup method before a downstream step requiring enzyme removal. For size-selected recovery, use M02. [3-5]

D. Check the digest

10. Run an aliquot with the expected fragment map and a size-appropriate ladder using M02. For troubleshooting, add a separate lane of the original undigested plasmid at comparable DNA mass, with loading dye but no digestion incubation. For a double digest, separate single-enzyme controls help identify which cut failed.

A digest made with 1× Green Buffer can be loaded directly. For colorless buffer, add loading dye at its final working dilution. Keep total loading dye at the intended concentration if the PCR mix already contains dye. Preserve remaining digest for cleanup or downstream use. [7]

p.s.

Reserve original plasmid before digestion. Comparing it with the digest is more informative than interpreting a broad band by shape alone.

Extend a confirmed partial digest only within the shortest enzyme limit. BglII/MluI-only reactions may be extended up to 16 h if needed; a digest containing FastDigest EcoRI should not be left overnight. [1, 3-5]

Do not keep adding enzyme to the same small tube. Recalculate the combined enzyme fraction and buffer concentration, or set up a fresh reaction. [4, 6]

Expected Outcome

Sequence-directed digestion gives fragments consistent with the full construct map, allowing for co-migration and small fragments outside the gel's detection range. A single linear band can be the correct result of a unique-site plasmid digest.

After DpnI treatment, the PCR product should remain intact while methylated parental DNA is cleaved. An unchanged PCR band alone neither proves failure nor confirms complete template removal.

Troubleshooting

Use M02 first for shared ladder, loading, running or imaging defects. The cases below address digestion-specific causes after those checks; band shape alone does not identify the cause.

Observed problem	Likely causes to check	Next action
Diffuse bands / smear	Overload or salt carryover; degraded DNA; star activity under unsuitable buffer, excess enzyme/glycerol or excessive time	Compare a smaller load in M02. If sample-specific, compare original DNA and an enzyme-free incubation control: existing degradation points to input quality; degradation during incubation suggests contamination. Remake with clean reagents and matched buffer. Shorten an overlong digest or reduce excess enzyme; more time can worsen star activity. [6, 8, 9]
More bands than predicted	Unaccounted-for recognition sites, partial digestion, plasmid conformations or unintended cleavage	Check whether each site is unique where intended across the complete sequence, including insert and junctions; predict all fragment sizes. Compare undigested and single-digest controls. If the map is correct, investigate partial digestion, star activity or contamination rather than assuming an enzyme has lost specificity. [8, 9]
Expected bands plus one extra broad band	Residual uncut or partially cut plasmid; overload or DNA-bound enzyme can also shift migration	On the next gel, add the original undigested plasmid in a separate lane. If the extra signal tracks uncut/partial material, extend within the enzyme window or repeat with fresh active enzyme and correct buffer. Overnight incubation is an option only when every enzyme permits it: BglII/MluI up to 16 h; EcoRI no longer than 30 min. [1, 3-5, 9]
Only one broad band	No digestion, or a correctly linearized plasmid whose band is broadened by loading	Check the predicted outcome first. If the sample resembles the uncut control, test enzyme activity on known cuttable DNA and verify FastDigest buffer, temperature, input purity and methylation sensitivity. A passing activity control points toward the sample or site context. [5, 6, 8]
Double digest resembles a single digest	One enzyme is inactive, inhibited or unable to cut near the newly created end	Run each enzyme separately on the same substrate. Check site sequence, flanking bases and methylation; MluI is blocked by CpG methylation. Correct the failing step or use sequential digestion with cleanup when a shared window is unsuitable. [1, 5, 8]
Fewer bands than expected	Similar-sized fragments co-migrate; a small fragment is faint or has run off; incomplete cutting	Compare predicted sizes with the ladder's useful range and M02 gel concentration. Improve separation before diagnosing enzyme failure. If a missing fragment is replaced by a larger sum-sized fragment, investigate partial digestion.
DpnI does not resolve suspected parental carryover	Unmethylated parent, absent methylated GATC sites, inactive enzyme or inhibitory PCR mixture	Check the parent's host and sites. Test DpnI on a known dam+ plasmid and compare a purified aliquot of the PCR material. Extra PCR amplicons require M01 Optimization O2, not additional DpnI. If only the target PCR band remains unchanged, that is expected. [2, 6]

Reference

[1] Thermo Fisher Scientific. Reaction Conditions for FastDigest Restriction Enzymes: substrate-specific times, end-flanking bases, heat inactivation and longest recommended incubation. Accessed 2026-09-16. Open source

[2] Thermo Scientific. FastDigest DpnI, FD1703, MAN0012574, Rev. 7, pp. 2-4. Methylation requirement and reaction setup. Accessed 2026-09-16. Open source

[3] Thermo Scientific. FastDigest EcoRI, FD0274, MAN0012438, Rev. 11, pp. 2-4. Substrate-specific digestion and 30-minute prolonged-incubation limit. Accessed 2026-09-16. Open source

[4] Thermo Scientific. FastDigest BglII, FD0084, MAN0012423, Rev. 9, pp. 2-4. Reaction setup, lack of thermal inactivation and extended-incubation assay. Accessed 2026-09-16. Open source

[5] Thermo Scientific. FastDigest MluI, FD0564, MAN0012470, Rev. 9, pp. 2-4. Conditions, CpG sensitivity and extended-incubation assay. Accessed 2026-09-16. Open source

[6] Thermo Fisher Scientific. FastDigest DpnI FAQs: crude-PCR buffer carryover, enzyme-volume limits, inhibitors and diagnostic controls. Accessed 2026-09-16. Open source

[7] Thermo Fisher Scientific. FastDigest DpnI product page: FastDigest Buffer B64 versus Green Buffer B71, direct loading and fluorescence compatibility. Accessed 2026-09-16. Open source

[8] Thermo Fisher Scientific. Restriction Enzyme Troubleshooting Guide: incomplete digestion, unexpected sites, diffused bands and end proximity. Accessed 2026-09-16. Open source

[9] Thermo Fisher Scientific. Troubleshooting Common Issues with Restriction Digestion Reactions: partial digestion, star activity, gel shift and unexpected recognition sites. Accessed 2026-09-16. Open source

[10] Thermo Fisher Scientific. DoubleDigest Calculator: buffer and reaction design for conventional Thermo restriction enzymes; FastDigest uses its universal buffer. Accessed 2026-09-16. Open source

[11] Sangon Biotech. B501005 DEPC-treated water, SDS, revision 2024-12-24; product identity. Accessed 2026-10-03. Open source